# A Literature-Implied Negative Answer to the Unrestricted Isotropic Log-Concave LSI Question

Run `fa_banach_001` — review packet

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**Status.** `literature_implied_answer (negative)`. The implication is standard and explicit in later literature; the application to the source question and the elementary Laplace certificate are recorded here. No originality is claimed.

## 1 Source question

Lee and Vempala ask whether the classical log-Sobolev constant is $\Theta(1)$ for every isotropic log-concave distribution [1, p. 21, immediately before Definition 36]. The exact source passage is reproduced below.

> **Tight log-Sobolev constant and improved concentration.** One can view the slicing conjecture as being weaker than the thin-shell conjecture and the thin shell conjecture as weaker than the KLS conjecture. Naturally one may ask if there is a conjecture stronger than the KLS conjecture. It is known that KLS conjecture is equivalent to proving Poincaré constant is $\Theta(1)$ for any isotropic logconcave distribution. It is also known that log-Sobolev constant (defined below) is stronger than the Poincaré constant. So, a natural question is whether the log-Sobolev constant is $\Theta(1)$ for any isotropic logconcave distribution? We first remind the reader of the definition.
>
> **Definition 36.** For any distribution $p$, we define the the log-Sobolev constant $\rho_p$ be the largest number such that for every smooth $f$ with $\int f^2(x)p(x)dx = 1$, we have that
>
> $$\int |\nabla f(x)|^2 p(x)dx \gtrsim \rho_p \int f^2(x) \log f(x)^2 \cdot p(x)dx.$$
>
> 21

In the source convention, after normalizing $\int f^2 \, d\mu = 1$, a positive constant means that

$$\int |\nabla f|^2 \, d\mu \geq c\rho_\mu \int f^2 \log(f^2) \, d\mu$$

for some absolute normalization constant $c > 0$ and every smooth $f$.

## 2 Literature identification

Bizeul states the standard implication that a classical log-Sobolev inequality forces all one-dimensional marginals to have subgaussian tails [2, p. 1, abstract and (1)–(4)]. Thus any isotropic log-concave distribution with a non-subgaussian marginal answers the source question negatively. The supporting passage is reproduced below.

# On the Log-Sobolev Constant of Log-Concave Vectors

Pierre Bizeul

**Abstract**

It is well known that if a random vector satisfies a log-Sobolev inequality, all of its marginals have subgaussian tails. In the spirit of the KLS conjecture, we investigate whether this implication can be reversed under a log-concavity assumption. In the general setting, we improve on a result of Bobkov, establishing the best dimension dependent bound on the log-Sobolev constant of subgaussian log-concave measures, and we investigate some special cases.

## 1 Introduction and results

A Borel probability measure $\mu$ on $\mathbb{R}^n$ is said to satisfy a logarithmic Sobolev inequality with constant $\rho > 0$ if for any smooth function $f : \mathbb{R}^n \to \mathbb{R}$, one has

$$\mathrm{Ent}_\mu(f^2) \leq 2\rho^2 \int_{\mathbb{R}^n} |\nabla f|^2 \, d\mu, \tag{1}$$

where for a nonnegative function $g$,

$$\mathrm{Ent}_\mu(g) = \mathbb{E}_\mu[g \log g] - \mathbb{E}_\mu[g] \log \mathbb{E}_\mu[g],$$

and $\mathbb{E}_\mu(h) = \int_{\mathbb{R}^n} h \, d\mu$ denotes expectation with respect to $\mu$. Here $|\cdot|$ denotes the Euclidean norm. We denote by $\rho_{LS}(\mu)$ the optimal constant $\rho$ such that (1) holds. It is well known that the log-Sobolev inequality implies Gaussian concentration: the Herbst argument yields a quadratic bound on the logarithmic Laplace transform of Lipschitz functions.

$$\log \int e^{sf} \, d\mu \ \leq \ \frac{\rho^2 s^2 \, |f|_{\mathrm{Lip}}^2}{2} \ + \ s \int_{\mathbb{R}^n} f \, d\mu, \qquad s \in \mathbb{R}, \tag{2}$$

where $|f|_{\mathrm{Lip}}$ is the Lipschitz constant of $f$. Markov's inequality then implies Gaussian concentration of $f$ around its mean,

$$\mu(|f - \mathbb{E}_\mu(f)| \geq t) \leq 2e^{-\frac{t^2}{2|f|_{\mathrm{Lip}}^2 \rho^2}}. \tag{3}$$

Taking $f$ to be a linear form, we deduce that every one-dimensional marginal of $\mu$ is subgaussian. Classically, the log-Laplace estimate (2) can be reformulated in a way that will be convenient for our purposes. Namely, if $f$ is a 1-Lipschitz function satisfying $\int f \, d\mu = 0$, then

$f^2$

The supporting authors do not cite or name the source question. Accordingly, this is an agent-identified implication, not an explicit claimed resolution in the literature.

## 3 Explicit Laplace certificate

Let $\alpha = \sqrt{2}$ and let $\mu$ have density

$$p_\alpha(x) = \frac{\alpha}{2} e^{-\alpha|x|}, \qquad x \in \mathbb{R}.$$

The function $\log p_\alpha$ is concave. Symmetry gives $\mathbb{E}_\mu X = 0$, and

$$\mathbb{E}_\mu X^2 = \frac{2}{\alpha^2} = 1,$$

so $\mu$ is isotropic and log-concave already in dimension one.

For $|s| < \alpha$, its moment-generating function is

$$M(s) = \mathbb{E}_\mu e^{sX} = \frac{\alpha^2}{\alpha^2 - s^2}.$$

Fix $0 < t < \alpha/2$ and define the smooth normalized function

$$f_t(x) = \frac{e^{tx}}{\sqrt{M(2t)}}.$$

Then $\int f_t^2 \, d\mu = 1$ and

$$\int |f_t'|^2 \, d\mu = t^2.$$

On the other hand,

$$\begin{aligned} \mathrm{Ent}_\mu(f_t^2) &= \int f_t^2 \log(f_t^2) \, d\mu \\ &= 2t \frac{M'(2t)}{M(2t)} - \log M(2t) \\ &= \frac{8t^2}{\alpha^2 - 4t^2} - \log\left(\frac{\alpha^2}{\alpha^2 - 4t^2}\right). \end{aligned}$$

As $t \uparrow \alpha/2$, this entropy tends to $+\infty$, while the Dirichlet energy tends to $\alpha^2/4 = 1/2$. Consequently

$$\inf_f \frac{\int |f'|^2 \, d\mu}{\mathrm{Ent}_\mu(f^2)} = 0,$$

and the source-normalized log-Sobolev constant is exactly zero. This proves the negative answer without invoking the general concentration implication.

# 4 Scope, search bounds, and review verdict

The conclusion applies to the unrestricted wording only. It does not contradict the bounded-support estimate in the source, nor does it decide dimension-free LSI bounds under a subgaussian-tail assumption.

The bounded novelty/status search covered the run's registry, solution, attempt, and proof-gap indexes; exact-title and exact-phrase searches; and arXiv:1712.01791 and arXiv:2306.12997. No source was found that explicitly claims to answer the Lee–Vempala question. Since arXiv:2306.12997 records the decisive standard implication, the correct classification is *literature-implied answer*, not an original counterexample.

**Human-review recommendation.** Accept the negative implication and the elementary certificate. Preserve the scoped literature-implied status.

# References


[1] Y. T. Lee and S. S. Vempala, *The Kannan–Lovász–Simonovits Conjecture*, arXiv:1807.03465 (2018).

[2] P. Bizeul, *On the Log-Sobolev Constant of Log-Concave Vectors*, arXiv:2306.12997v3 (2026), accepted in the Journal of Functional Analysis.